BIOLOGY EXAM — MARKING REPORT

SSC-I (9th Grade) — Annual Examination 2026 — Version 3

Assessment Date: 3 April 2026 | Report Updated: 4 April 2026

Student: Seemab	**Subject:** Biology	**Paper:** Version 3	**Total Marks:** 65

TOTAL SCORE

52 / 65

80% — Grade A

Section	Score	Maximum	Percentage
A — MCQs (12 questions)	**12**	12	**100%**
B — Short Questions (all 11)	**28**	33	**84.8%**
C — Long Questions (4 questions)	**12**	20	**60%**
TOTAL	**52**	**65**	**80%**

SECTION A — Multiple Choice Questions [12 × 1 = 12 Marks]

#	Question	Correct	Seemab	Mark
1	The study of internal structure of organisms is called	B) Anatomy	B	**1**
2	The five-kingdom classification system was introduced by	D) Robert Whittaker	D	**1**
3	Organisms that feed on dead, decaying matter are called	C) Saprotrophs	C	**1**
4	The organelle responsible for intracellular digestion is	C) Lysosome	C	**1**
5	The cell wall of plant cells is mainly made up of	C) Cellulose	C	**1**
6	The shape of red blood cells is	D) Biconcave	D	**1**
7	The study of distribution of plants and animals in different geographical regions	B) Biogeography	B	**1**
8	The monomer unit of proteins is	B) Amino acid	B	**1**
9	Viruses are not included in any kingdom because	C) Not considered organisms	C	**1**
10	Lactose is an example of	B) Disaccharide	B	**1**
11	Which organelle is called the powerhouse of the cell?	C) Mitochondria	C	**1**
12	Carolus Linnaeus introduced the naming system known as	C) Binomial nomenclature	C	**1**

Section A total: 12/12 (100%) — Perfect score.

SECTION B — Short Answer Questions [11 × 3 = 33 Marks]

Q#	Topic	Score	Assessment
(i)	Biology definition + 3 divisions	3/3	**Excellent** Complete definition with Greek roots (bios, logos). Botany, Zoology, Microbiology all correct with relevant examples (photosynthesis, nervous system of frogs, bacteria in yogurt fermentation).
(ii)	Interdisciplinary research collaboration	3/3	**Excellent** Good definition. Biophysics (muscles, nerve impulses, X-ray diffraction/DNA) and Biochemistry (chemical reactions in organisms, protein/nucleic acid structure) are valid textbook examples.
(iii) OR	Four kingdoms of Domain Eukarya	2/3	**Needs detail** All four kingdoms named correctly (Protista, Animalia, Plantae, Fungi). Descriptions are too brief. Missing: specific examples (Amoeba, Paramecium, mushroom, yeast, moss, insects), nutrition modes for each kingdom.
(iv) OR	Prokaryotic vs Eukaryotic cells	3/3	**Excellent** Comprehensive comparison covering nucleus, membrane-bound organelles, size (1–10μm), DNA shape (circular vs linear), ribosomes (70S vs 80S), and examples (bacteria/archaea vs plant/animal cells). Well structured.
(v)	Mesophyll vs Epidermal cells	2.5/3	**Good** Location, chloroplasts, and function compared well. Missing: specialised epidermal cells — **guard cells** (stomata) and **root hair cells** (absorption).

(vi) OR	Define tissue + 4 animal tissue types	3/3	**Excellent** Clear definition. All four types (epithelial, muscle, nerve, connective) described with structure, function, types/examples. Mentioned actin & myosin in muscle tissue. Well organised answer.
(vii)	Nucleic acids, purines & pyrimidines	2/3	**Error** Definition and examples correct. Ring structures section has cross-outs showing confusion. Size section correctly states purines = larger (double ring), pyrimidines = smaller (single ring), but the direct comparison had corrections/contradictions. Missing base-pairing rules (A–T, G–C).
(viii)	Three types of carbohydrates	2/3	**Error** Content is correct but **column headers swapped** — polysaccharide definition/formula/examples written under disaccharide heading and vice versa. Formulae and examples match the (wrong) headings consistently. Missing sucrose (most important disaccharide).
(ix)	Four careers in biology	3/3	**Excellent** Four careers listed (Biotechnology, Fisheries, Medicine/Surgery, Horticulture). Biotechnology and Medicine described in good detail with correct information.
(x) OR	Define data, hypothesis, vector	2.5/3	**Good** All three definitions correct (data = facts/observations, hypothesis = tentative/testable/falsifiable, vector = transmitter organism). Missing specific examples: female *Anopheles* mosquito, *Aedes aegypti*.
(xi) OR	Binomial nomenclature	2/3	**Needs detail** Correct: Carolus Linnaeus, two Latin names, genus (capital) + species (lowercase). Examples (*Homo sapiens*, *Solanum tuberosum*) good. Missing: Linnaeus dates (1707–1778), importance (universal system, avoids confusion), italicisation rule.

Section B total: 3 + 3 + 2 + 3 + 2.5 + 3 + 2 + 2 + 3 + 2.5 + 2 = 28/33 (84.8%) — Strong performance across all 11 questions.

SECTION C — Detailed Answer Questions [4 × 5 = 20 Marks]

Q.3 OR — Biological Method / Discovery of Malaria Cause **4.5 / 5**

Strong Very thorough answer covering all steps of the biological method:

✔ Recognition of problem (malaria since ancient times, cause unknown)
✔ Observations (4 key points: marshy areas, drinking water, quinine, Plasmodium in blood)
✔ Hypothesis (Laveran, 1882: "Plasmodium is the cause of malaria")
✔ Deduction ("If Plasmodium is the cause, all ill persons should have it in blood")
✔ Experiment (100 patients + 100 healthy control group)
✔ Results (almost all patients had Plasmodium, 7 healthy had it — incubation period, later became ill)
✔ Conclusion ("Results were convincing to prove the hypothesis")
✔ Reporting (results announced worldwide)

✘ **Missing:** Spread of malaria section — A.F.A. King (1883) suggested mosquitoes spread malaria, **Ronald Ross (1897)** proved female *Anopheles* mosquito transmits malaria. This is worth ~0.5 marks.

Q.4 OR — Proteins: Composition, Structure & Functions **3.5 / 5**

Good Solid answer with some gaps:

✔ Chemical composition (C, H, O, N, sometimes P, S, Fe) — correct
✔ Amino acid structure (alpha carbon, 4 groups: –H, $–NH_2$, –COOH, –R group) — detailed and correct
✔ 20–25 types of amino acids, R-group determines properties — correct
✔ Peptide bonds between amino acids — correct
✔ Four structural levels named: primary, secondary, tertiary, quaternary
✔ Functions: antibodies, histones, enzymes — valid

✘ **Secondary structure:** Must include **alpha helix** (coiled spring shape) and **beta pleated sheet** (zigzag folding) — these are key terms examiners look for. Only wrote "folding with extra bonds".
✘ **Functions:** The expected textbook answers were: (1) **Structural** (collagen in skin/bones, keratin in hair/nails), (2) **Enzymatic** (amylase breaks starch), (3) **Transport** (haemoglobin carries O_2).

Q.5 — NOT ATTEMPTED **0 / 5**

Not Attempted This question was not answered. The exam requires all 4 Section C questions to be attempted.

Available options were:
• Q.5: Levels of biological organization (atomic → organism) with examples and complexity
• Q.5 OR: Classification of carbohydrates with examples, formulae, and sources

Revision note: Both options cover Ch 5 (Tissues/Organs) and Ch 6 (Molecular Biology) respectively — topics Seemab scored well on in Section B. This was likely a time management issue rather than a knowledge gap.

Q.6 OR — RNA Types, Central Dogma, Transcription & Translation **4 / 5**

Strong Comprehensive answer:

✔ mRNA (carries message from DNA in nucleus to cytoplasm for protein synthesis)
✔ tRNA (transfers amino acids to ribosomes during protein synthesis)
✔ rRNA (component of ribosomes, combines with ribosomal proteins — "factory of proteins")
✔ Central dogma: DNA → RNA → Protein (flow of biological information)
✔ Transcription: DNA strands unwind, one strand (coding strand) copied into mRNA, strands rejoin
✔ Translation: mRNA moves to cytoplasm, binds to ribosome, tRNA brings amino acids as per mRNA code, polypeptide chain formed
✔ Gene expression = transcription + translation

✘ **Missing:** RNA polymerase enzyme and start/stop codons for full marks.

Section C total: 4.5 + 3.5 + 0 + 4 = 12/20 (60%) — Q.5 not attempted cost 5 marks.

KEY MISTAKES TO REVIEW

1. Q.5 Not Attempted (Section C) — Lost 5 marks
This is the biggest mark loss. Seemab scored well on the related topics in Section B, so this was likely a **time management** issue. Practice completing all questions within the time limit.

2. Purines vs Pyrimidines (Q.vii) — Ring structures confused
✘ Cross-outs show confusion in the ring structure section
✔ **Correct:** Purines = DOUBLE ring (Adenine, Guanine) = larger. Pyrimidines = SINGLE ring (Cytosine, Thymine, Uracil) = smaller.
Memory trick: Purines are Pure and big (double ring). Pyrimidines are tiny (single ring).

3. Carbohydrates Table (Q.viii) — Column headers swapped
✘ Polysaccharide definition written under "Disaccharides" column and vice versa
✔ **Remember:** Di = two (2 monosaccharides, $C_{12}H_{22}O_{11}$, e.g. sucrose, lactose, maltose). Poly = many (long chains, $(C_6H_{10}O_5)n$, e.g. starch, cellulose, glycogen).
Also remember sucrose (glucose + fructose) = most common disaccharide (table sugar).

4. Protein Secondary Structure (Q.4 OR) — Missing key terminology
✔ Must mention: (1) **Alpha helix** = coiled spring shape, stabilised by hydrogen bonds. (2) **Beta pleated sheet** = zigzag/folded shape, also stabilised by hydrogen bonds.

5. Missing Specific Examples Throughout
✔ Key examples to memorise:
• Protista: *Amoeba, Paramecium*, algae • Fungi: mushroom, yeast, bread mold
• Vectors: female *Anopheles* mosquito (malaria), *Aedes aegypti* (dengue)
• Linnaeus: 1707–1778, Swedish botanist • Protein functions: collagen, keratin, haemoglobin

OVERALL ASSESSMENT

Strengths

✔ Perfect MCQ score (12/12) — strong conceptual understanding

✔ Section B performance is excellent (84.8%) — well-structured answers

✔ Q3 OR (biological method) was outstanding — covered all steps systematically

✔ Good use of tables and comparisons (prokaryotic/eukaryotic, mesophyll/epidermal)

✔ Q6 OR (RNA/central dogma) shows strong molecular biology understanding

✔ All 11 Section B questions attempted with solid answers

Areas for Improvement

✘ **Time management:** Q.5 in Section C was not attempted — lost 5 easy marks

✘ Double-check table headers and column labels before moving on

✘ Memorise specific textbook facts: dates, scientist names, examples

✘ Learn key terminology precisely (alpha helix, beta pleated sheet, base-pairing rules)

✘ Always include 2–3 specific examples for each point in long questions

Potential with corrections: If Q.5 had been attempted (estimated 3–4/5 based on Seemab's strong knowledge of these topics) and the minor errors corrected, Seemab could score **57–60 / 65 (88–92%) — a strong Grade A+**. The main priority is time management to ensure all questions are completed.